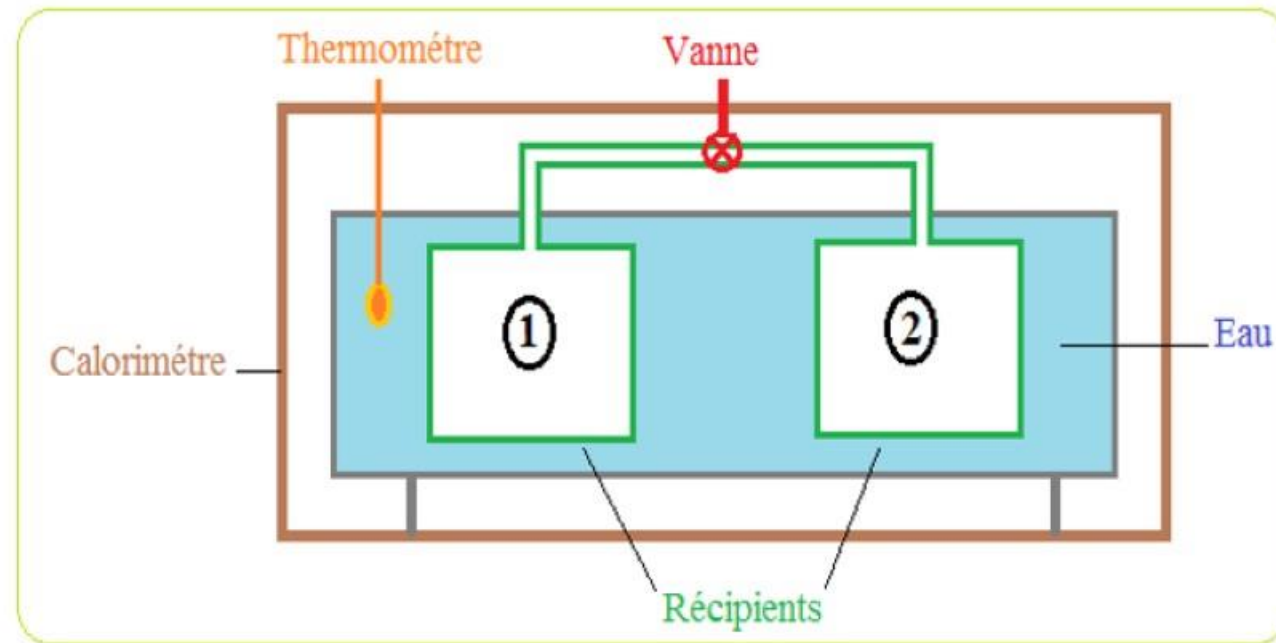


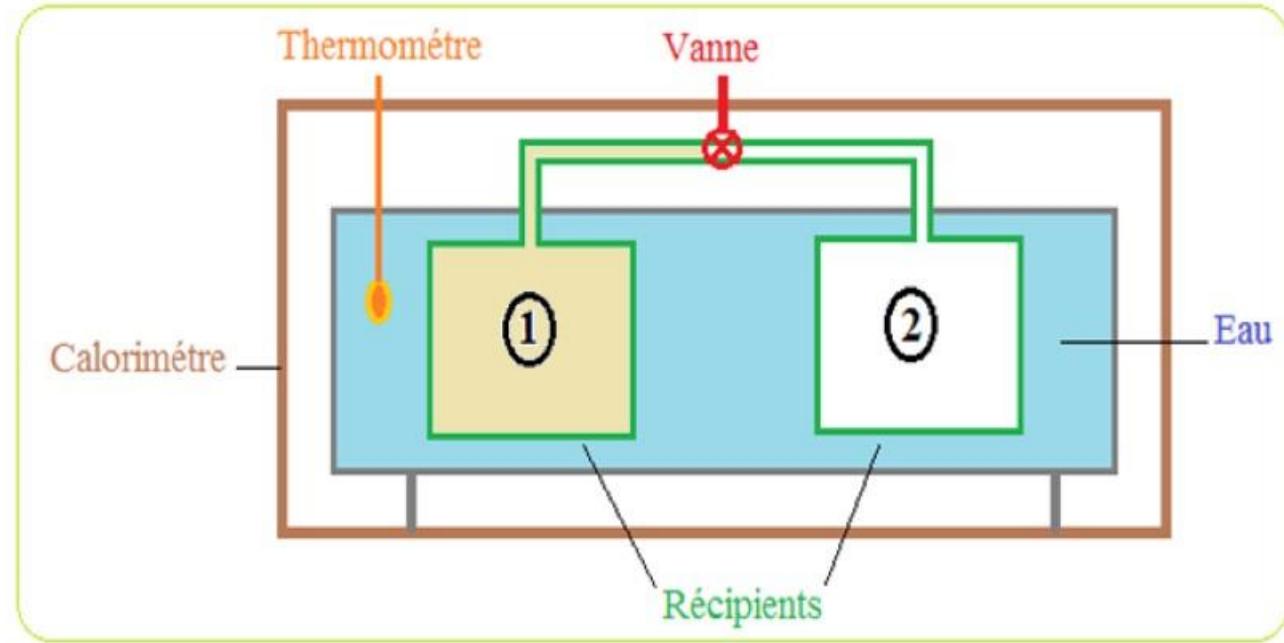
Lois de Joule

Première loi de Joule



Instant initial

$T = \text{cste}$



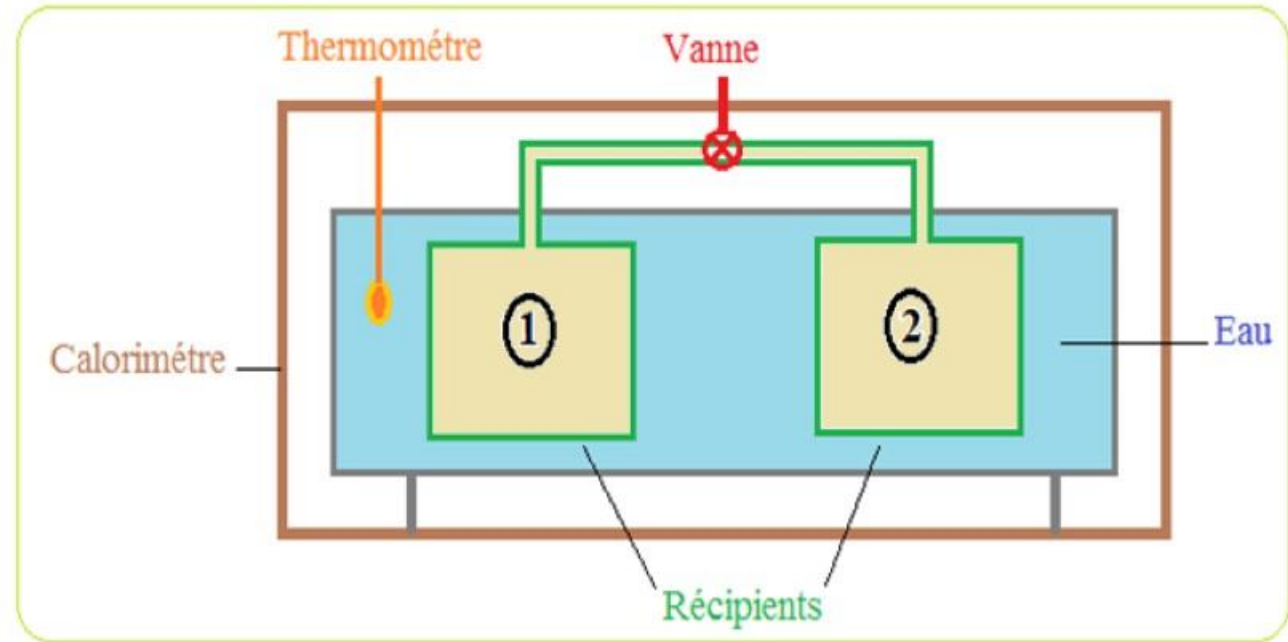
$$V_{\text{gaz}} = V_1 = \text{cste} \Rightarrow \delta W = -P_{\text{ext}} dV = 0 \Rightarrow W = 0$$

$$\text{Calorimètre (Système adiabatique)} \Rightarrow Q = 0$$

$$\text{Premier principe} \Rightarrow \Delta U = W + Q = 0$$

Instant final

$T = \text{cste}$



$$V_{\text{gaz}} = V_1 + V_2 = \text{cste} \Rightarrow \delta W = -P_{\text{ext}} dV = 0 \Rightarrow W = 0$$

Calorimètre (Système adiabatique) $\Rightarrow Q = 0$

Premier principe $\Rightarrow \Delta U = W + Q = 0$

$$V_{1\text{gaz}} = V_1 \rightarrow \Delta U = 0$$

$$V_{2\text{gaz}} = V_1 + V_2 \rightarrow \Delta U = 0$$

'U' ne dépend pas de 'V'

$$\left(\frac{\partial U}{\partial V}\right)_T = 0$$

Formule de ΔU

$$U = f(V, T)$$

$$dU = \left(\frac{\partial U}{\partial V}\right)_T dV + \left(\frac{\partial U}{\partial T}\right)_V dT$$

$$\text{mais } \left(\frac{\partial U}{\partial V}\right)_T = 0 \Rightarrow dU = \left(\frac{\partial U}{\partial T}\right)_V dT$$

$$V = \text{cste} \Rightarrow \delta W = -P_{\text{ext}} dV = 0$$

$$dU = \delta W + \delta Q_V \Rightarrow dU = \delta Q_V$$

$$dU = \delta Q_V \frac{dT}{dT} \Leftrightarrow dU = \frac{\delta Q_V}{dT} dT$$

$$\Rightarrow dU = C_V dT$$

$$\left. \begin{aligned} dU &= C_V dT \\ C_V &= m c_v = n \dot{c}_v \end{aligned} \right\} \begin{cases} dU = m c_v dT \\ dU = n \dot{c}_v dT \end{cases}$$

$$dU = \delta Q_v \Rightarrow \Delta U = Q_v = \int C_V dT$$

$$C_p - C_v = nR$$

Deuxième loi de Joule

$$H = U + PV \Rightarrow dH = dU + d(PV)$$

$$\left. \begin{array}{l} PV = nRT \\ dU = C_v dT \end{array} \right\} dH = C_v dT + d(nRT)$$

$$dH = C_v dT + nR dT$$

$$dH = (C_v + nR) dT$$

$$dH = C_p dT$$

The diagram illustrates the derivation of specific heat capacity formulas. On the left, two equations are grouped by a large green curly brace:

$$dH = C_p dT$$
$$C_p = m c_p = n \dot{c}_p$$

Two blue arrows originate from the center of the brace and point to the right, leading to two separate equations:

$$dH = m c_p dT$$
$$dH = n \dot{c}_p dT$$

$$H = f(P, T)$$

$$dH = \left(\frac{\partial H}{\partial P}\right)_T dP + \left(\frac{\partial H}{\partial T}\right)_P dT$$

$$dH = c_p dT$$

$$\left(\frac{\partial H}{\partial P}\right)_T = 0 \quad \text{et} \quad \left(\frac{\partial H}{\partial T}\right)_P = c_p$$

$$c_p = \frac{\delta Q_p}{dT}$$

$$dH = \delta Q_p \Rightarrow \Delta H = Q_p = \int c_p dT$$

$T = \text{cste}$

$$dT = 0 \Rightarrow dU = C_V dT = 0$$

$$\Rightarrow \Delta U = 0$$

$$\Rightarrow U = \text{cste}$$

$T = \text{cste}$

$$dT = 0 \Rightarrow dH = C_P dT = 0$$

$$\Rightarrow \Delta H = 0$$

$$\Rightarrow H = \text{cste}$$

Application du 1^{er} principe sur différentes transformations

$$\Delta U = W + Q$$

$$V = cste \Rightarrow Q = Q_V = \Delta U$$

$$\Delta U = \int C_V dT$$

$$P = cste \Rightarrow Q = Q_P = \Delta H$$

$$\Delta H = \int C_P dT$$

$$T = cste \Rightarrow \Delta U = 0 \text{ et } \Delta H = 0$$

$$\Rightarrow Q = -W$$

ΔU

ΔH

W

Q

Transformation isobare

$$P = cste \Rightarrow W = - \int P_{ext} dV = - \int P dV = - P \int dV = - P \Delta V$$

$$P = cste \Rightarrow Q = Q_p = \Delta H = \int C_p dT = C_p \int dT = C_p \Delta T$$

$$\Delta U = W + Q = W + Q_p = C_p \Delta T - P \Delta V$$

Transformation isochore

$$V = cste \Rightarrow dV = 0 \Rightarrow \delta W = -P_{ext} dV = 0 \Rightarrow W = 0$$

$$V = cste \Rightarrow Q = Q_V = \Delta U = \int C_V dT = C_V \int dT = C_V \Delta T$$

$$\Delta H = \int C_P dT = C_P \int dT = C_P \Delta T$$

Transformation isotherme

$$\Delta H = \int C_p dT = C_p \int dT = C_p \Delta T$$

$$T = cste \Rightarrow \Delta T = 0 \Rightarrow \Delta H = 0$$

$$\Delta U = \int C_v dT = C_v \int dT = C_v \Delta T$$

$$T = cste \Rightarrow \Delta T = 0 \Rightarrow \Delta U = 0$$

$$\Rightarrow W = -Q$$

Transformation réversible

Transformation irréversible

Transformation isotherme

Transformation irréversible

$$P_{ext} = P_f$$

$$W = - \int P_{ext} dV$$

$$W = - \int P_f dV$$

$$W = - P_f \int dV$$

$$\Rightarrow W = -P_f \Delta V \quad \Rightarrow Q = P_f \Delta V$$

Transformation isotherme

Transformation réversible $P_{ext} = P_g = \frac{nRT}{V}$

$$W = - \int P_{ext} dV$$

$$\Rightarrow W = - \int_1^2 P_g dV = - \int_1^2 \frac{nRT}{V} dV$$

$$\Rightarrow W = -nRT \int_1^2 \frac{dV}{V}$$

$$W = -nRT [\ln V]_1^2 = -nRT (\ln V_2 - \ln V_1)$$

$$\Rightarrow W = -nRT \ln \frac{V_2}{V_1} = nRT \ln \frac{V_1}{V_2} \quad \Rightarrow Q = nRT \ln \frac{V_2}{V_1} = -nRT \ln \frac{V_1}{V_2}$$

$$W = nRT \ln \frac{V_1}{V_2}$$

$$Q = nRT \ln \frac{V_2}{V_1}$$

Transformation isotherme

Loi de Mariotte

$$PV = cste \Rightarrow P_1 V_1 = P_2 V_2 \Rightarrow \frac{V_1}{V_2} = \frac{P_2}{P_1}$$

$$\Rightarrow W = nRT \ln \frac{P_2}{P_1}$$

$$\Rightarrow Q = nRT \ln \frac{P_1}{P_2}$$

Transformation adiabatique

$$\delta Q = 0$$

$$dU = \delta W + \delta Q$$

$$\Rightarrow dU = \delta W$$

$$C_V dT = -PdV$$

$$C_V dT + PdV = 0$$

$$dT = -\frac{P}{C_V} dV$$

1

2

$$PV = nRT \longrightarrow d(PV) = d(nRT)$$

$$PdV + VdP = nRdT$$

$$\Rightarrow PdV = nRdT - VdP$$

3

$$C_V dT + nRdT - VdP = 0$$

$$(C_V + nR)dT - VdP = 0$$

$$C_p dT - VdP = 0$$

4

$$-C_p \frac{P}{C_v} dV - V dP = 0$$

$$\Rightarrow -\gamma P dV - V dP = 0$$

$$\Rightarrow \gamma P dV + V dP = 0$$

÷ PV

$$\gamma \frac{dV}{V} + \frac{dP}{P} = 0$$

∫

$$\ln V^\gamma + \ln P = cste$$

$$\ln P V^\gamma = cste$$

$$PV^\gamma = cste$$

Transformation adiabatique

$$\left. \begin{array}{l} Q = 0 \\ \Delta U = W + Q \end{array} \right\} \Rightarrow \Delta U = W$$

1^{ère} Méthode $\rightarrow \Delta U = \int C_V dT \rightarrow W = \Delta U$

2nd Méthode $\rightarrow W = - \int P_{ext} dV \rightarrow \Delta U = W$

$$\Delta H = \int C_P dT$$

Transformation adiabatique

Formule de W

$$\delta W = -P dV = (-P dV) \frac{V^\gamma}{V^\gamma}$$

$$\delta W = -P V^\gamma \frac{dV}{V^\gamma} \Rightarrow W = - \int_1^2 P V^\gamma \frac{dV}{V^\gamma}$$

$$PV^\gamma = cste \Rightarrow W = -P V^\gamma \int_1^2 \frac{dV}{V^\gamma}$$

$$\Rightarrow W = -P V^\gamma \left[\frac{V^{-\gamma+1}}{-\gamma+1} \right]_1^2$$

$$\Rightarrow W = \left[-P V^\gamma \frac{V^{-\gamma+1}}{-\gamma+1} \right]_1^2 = \left[\frac{PV}{\gamma-1} \right]_1^2$$

$$\Rightarrow W = \frac{P_2 V_2 - P_1 V_1}{\gamma - 1}$$